

Live API Data Integration: Real-Time Dashboards with Python & Power BI

Project Questions & Objectives

This project focuses on building a live, interactive dashboard that shows important global statistics such as health, education, employment, and other key indicators. The data comes from a public API that is regularly updated with the latest information. Using Python, we write a script that automatically fetches this data from the API and processes it for analysis. This ensures that the data is fresh and accurate without manual downloads or updates.

The Python script is then connected directly to Power BI, creating a live data connection. Because of this live integration, the Power BI dashboard refreshes automatically whenever new data is available. Users do not need to perform any manual steps to update the dashboard, making it highly efficient and reliable for ongoing monitoring. The dashboard provides easy-to-understand visuals like charts and graphs, which help users quickly grasp trends and comparisons across countries and regions.

Once the connection is created, the dashboard will answer the following questions:

1. What is the average economic and development status of countries?

Find the average:

- GDP per capita
- Trade value
- Health expenditure
- GDP growth
- Forest area

2. Which regions spend more on healthcare?

Compare the average health expenditure across different regions.

3. How have important indicators changed over time?

Analyze the trends of:

- GDP
- Internet users
- Mobile subscriptions
- Forest area
- Renewable energy
- Unemployment

4. Does internet access improve immunization?

Check whether countries with more internet users also have higher immunization rates.

5. Does internet access help reduce unemployment?

Check whether countries with more internet access have lower unemployment.

6. Which countries have the highest average life expectancy?

Show the Top 10 countries by average life expectancy.

7. How are health indicators related?

Find the relationship between:

- Health expenditure
- Life expectancy
- Immunization
- Child mortality
- Disease burden

8. Does higher health spending increase life expectancy?

Compare health expenditure and life expectancy using a trend line or scatter chart to see whether higher spending is associated with longer life expectancy.